A Related Works

Conversational Recommendation: Conversational recommendation systems provide recommendations through natural language dialog with users [18, 10, 51]. Traditional approaches use separate modules for dialogue and recommendation [51, 9, 54], resulting in a complex pipeline. Recent research explores using LLMs to unify these components. Early studies investigated zero-shot recommendation capabilities through prompt engineering [20, 31], demonstrating LLMs could generate reasonable recommendations without specific training. However, a key challenge is the semantic gap between LLMs and recommender systems, as items are typically indexed by unique identifiers outside the LLM's vocabulary [23], making direct application difficult. More recent approaches address this by fine-tuning LLMs on recommendation data [53, 50, 17], enabling them to learn associations between natural language queries and item representations.

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Generative Retrieval: Generative retrieval reformulates traditional retrieval as a generation problem, directly generating item identifiers instead of using similarity matching. This approach offers significant advantages over traditional approaches, including constant computational complexity regardless of database size and improved performance through structured item identifier generation. Differentiable Search Index (DSI) [43] pioneered generative retrieval, predicting document identifiers directly from queries using transformers without explicit similarity computations. Neural Corpus Indexer (NCI) [46] built on this, generates document identifiers directly from queries using hierarchical semantic encoding and a prefix-aware weight-adaptive decoder. Recently, Transformer Index for Generative Recommenders (TIGER) [39] developed a generative retrieval approach for recommendation similar to the proposed system, directly generating content-based item tokens using semantic ID generation through content feature encoding and Residual Vector Quantization (RVQ).

Music Retrieval and Recommendation: Music recommendation systems face unique challenges due to complex musical features and user behaviors [41]. Most of the traditional approaches fall into three categories: collaborative filtering [22], content-based methods [45, 27, 36], and hybrid approaches combining both strategies. Natural language-based music retrieval has emerged to address limitations of traditional recommendation systems, enabling users to express preferences through text [33, 24, 15, 14]. Conversational music recommendation systems extend this approach through multi-turn dialogues for more interactive discovery [7, 12]. However, these systems typically separate dialogue management from recommendation components, using language models as an human interface. Developing a unified solution that learns the end-to-end connection of user queries, recommendations, system response in a multi-turn manner remains an open challenge in the field.

To address these challenges, TALKPLAY employs an LLM as a conversational music recommendation engine for both item retrieval and the dialogue management. While we propose a generative retrieval approach similar to [37], our research differs significantly by being an LLM-based conversational system that incorporates a multimodal tokenizer combining not only metadata but also audio features, lyrics, playlist co-occurrence information, and semantic tokens.

B Synthetic Data

B.1 Prompt to Google Gemini to synthesize conversation

```
You are an AI assistant specializing in music. Generate an ANSWER to the QUESTION in
     valid LIST OF JSON:
- role: one of {"user", "assistant", "music"}
- content:
```

```
  - for "user": string of conversation text for music request
  - for "assistant": string of conversation text for answer to user's request.
  - for "music": string of unique track_id (i.e., "3pzjHKrQSvXGHQ98dx18HI", "7
    vFv0yFGMJW3qVXbAd9BK9")

1. Conversation Start:
- The conversation begins with a user query.
  - This query can be a specific music item request (i.e., "superstition by steve
    wonder")
  - This query can be a natural language request (i.e., "party anthem with energetic
     rhythm", "play some 90s rock music")
- The assistant should respond with two consecutive objects:
  - A text response (role: "assistant") providing a brief reason for the
    recommendation
  - A music recommendation (role: "music") containing the `track_id`

2. Assistant Recommendations:
 - The assistant should recommend just one song per single turn, one that shares
    common musical traits (i.e, genre, artist, mood, theme, tempo) with the query
    song.
 - The goal is to syntehsize an engaging conversation and allow the user to explore
    music based on their request.
 - The assistant's music recommendations prioritize music tracks and tags, and
    ignore non-musical words. If there are captions and lyrics, use them as
    optional information as needed.

3. User Satisfaction:
 - In each turn, the user may or may not be satisfied with the recommendations and
    simply listen.
 - The user may express agreement with the recommendation (e.g., "This is great!").

4. User Direction Change:
 - In the real-world scenario, sometimes users are not happy with the recommendation
    . We need to include it in our synthetic conversation.
 - In such a case, the user would express dissatisfaction with a recommendation.
 - The user may specify a new direction or trend of the recommended songs. (e.g., "
    Play something different", "Play something more upbeat").
 - Use the songs in the "NEGATIVE SET" to synthesize this recommendation that user
    does not like.
 - This helps evaluate how the assistant responds to negative feedback and adapts
    its recommendations.

5. assistant Adaptation:
The assistant should provide the next music track based on the following factors:
 - The previous conversation.
 - The current music being played.
 - The user's recent requests.
 - The assistant should recommend at least half of the songs in the provided
    playlist.

7. JSON Format:
 - The conversation should be formatted as a list of JSON objects.

8. Example:
QUESTION :
Music Playlist:
{'5tkDCf3xKRZjT5RbCb3sZB': "Metadata:\n  Title: Big Amount\n  Artist: 2 Chainz,
    Drake\n  Album: Pretty Girls Like Trap Music, Industry Kings 2\nTag: drake, Hip-
    Hop/Rap, rap\n  Popularity: 55.0\nLyrics:  Pat downs, [REST OF THE LYRICS ARE
    OMITTED]\nPseudo Caption: This music features a hip hop song that consists of a
     flat male vocal rapping over shimmering hi hats, claps, punchy kick, groovy
    bass, synth lead melody and tinny hi hats. It sounds energetic, exciting and
    addictive - like something you would hear in clubs.\nTempo Estimate: 134.13\
    nKey Estimate: G# minor",
[REST OF THE TRACKS ARE OMITTED]}
```

```
Negative Set:
{'7bJ4mu7MHa3rHiNyKjOoSl': 'Metadata:\n  Title: Ice Melts\n  Artist: Young Thug,
    Drake\n  Album: More Life\nTag: young thug, Hip-Hop/Rap\n  Popularity: 60.0\
    nLyrics:  You need to give it up to someone. [REST OF THE LYRICS ARE OMITTED]\
    nPseudo Caption: This music features a hip hop song that consists of a
    passionate male vocal, alongside harmonizing background vocals, singing over
    shimmering hi hats, punchy kick and claps. It sounds groovy and addictive -
    like something you would hear in clubs.\nTempo Estimate: 124.51\nKey Estimate:
    A major',
[REST OF THE TRACKS ARE OMITTED]}

ANSWER:
[
  {"role": "user","content": "Play something like 'Big Amount' by 2 Chainz and Drake
    "},
  {"role": "assistant","content": "Here's a similar track with a similar vibe,
    featuring a catchy hook and hard-hitting flow."},
  {"role": "music","content": "4ckuS4Nj4FZ7i3Def3Br8W"},
  {"role": "user","content": "This is great! Can you play something more upbeat, but
     still in a hip hop style?"},
  {"role": "assistant","content": "Here's a track with a faster tempo and energetic
    rhythm, perfect for keeping the party going."},
  {"role": "music","content": "0B8QzDH7YWih85V5SEMnyJ"},
  {"role": "user","content": "I'm not feeling this one, play something different"},
  {"role": "assistant","content": "I understand. How about something with a bit more
     of a soulful feel, but still with a good groove?"},
  {"role": "music","content": "0PXp9QxBfSKGGpWUNaws2X"},
  {"role": "user","content": "Yeah, that's better, this is more my style"},
  {"role": "assistant","content": "Great choice! Let's keep it in this vein, with a
    focus on smooth vocals and melodic rhythms."},
  {"role": "music","content": "0H8XeaJunhvpBdBFIYi6Sh"},
  {"role": "user","content": "This is so good, this mood is perfect"},
  {"role": "assistant","content": "I'm glad you're enjoying it! Here's another track
     that leans into the same kind of soulful vibe."},
  {"role": "music","content": "3LwYGYpsg5SQioMX5BwFAz"},
  {"role": "user","content": "This is really good! One last request, something a
    little more intense with a little more aggression"},
  {"role": "assistant","content": "You got it. Here's a track with a harder edge and
     more energetic production. Get ready to bump!"},
  {"role": "music","content": "0utlOiJy2weVl9WTkcEWHy"}
]
```

B.2 Synthetic Conversation Example

This is one of the synthetic conversation results.

```
[{"role": "user", "content": "I'm looking for some mellow acoustic
    tracks to relax to. Can you suggest something?"},
 {"role": "music", "content": "3kZC0ZmFWrEHdUCmUqlvgZ"},
 {"role": "assistant", "content": "I think I Will Follow You into the
     Dark by Death Cab for Cutie would suit your mood. It's known for
     its emotional depth and gentle acoustic sound."},
 {"role": "user", "content": "That was beautiful, but a little too
     somber.  How about something with more energy, still in the
     alternative rock genre?"},
 {"role": "music", "content": "3mcG2NI5G5vhrQtRda1YnA"},
 {"role": "assistant", "content": "Okay, let's try something more
     upbeat. The Middle by Jimmy Eat World is a popular alternative
     rock anthem that offers a good balance of energy and emotion."}]
```

C A-vs-B Human Evaluation Interface

The Survey Interface is a web-based evaluation tool designed to allow participants to compare outputs from two models (A and B). For each evaluation item, users are presented with the same input prompt along with two model responses shown in random order. Participants are asked to choose the better response based on specific criteria—such as Recommendation Relevance and Response Naturalness—or select "Tie".

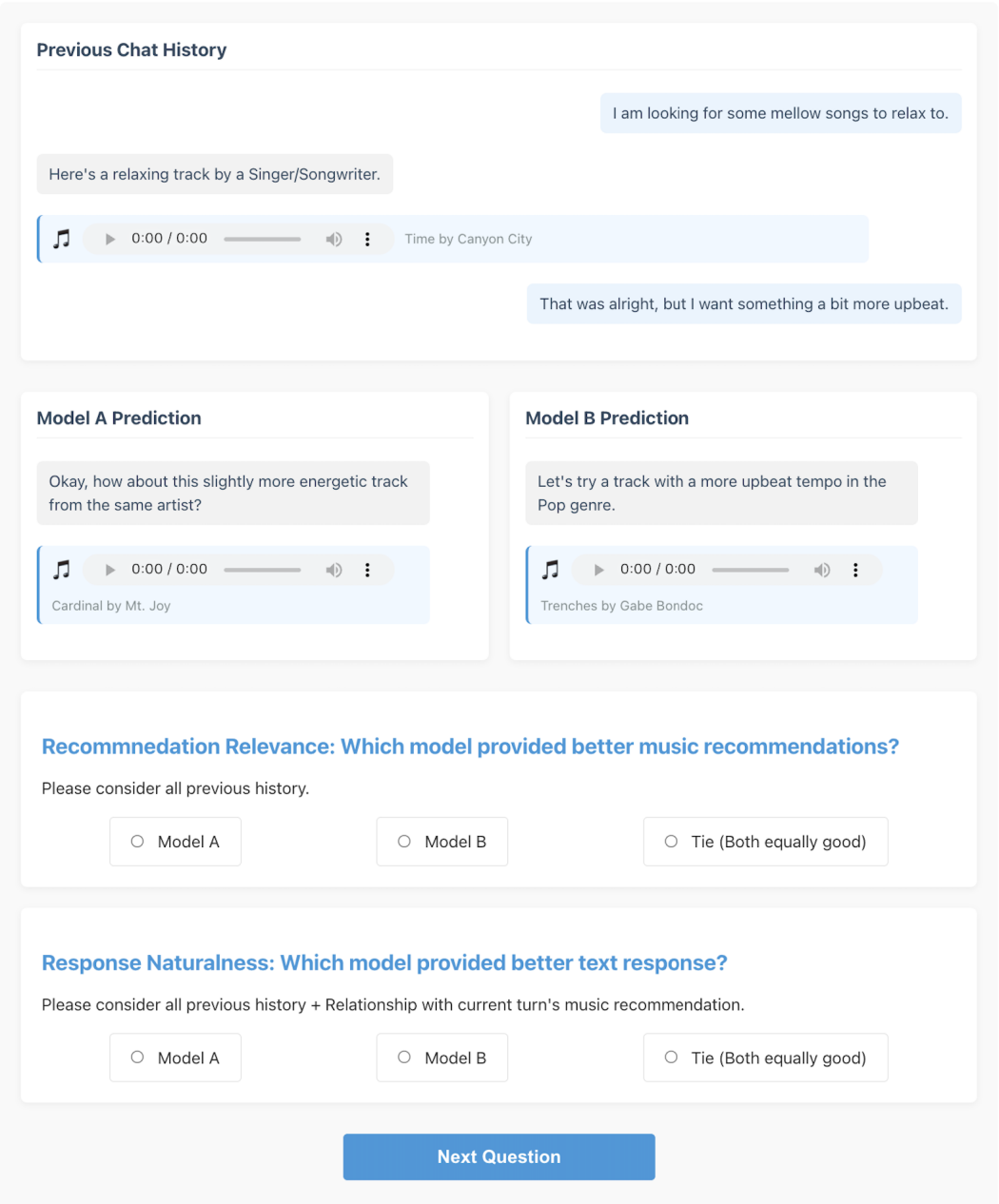

Figure 4: Survey Interface of Subjective Evaluation.